

Automatic Lab Allocation System

Motivation

Traditional lab scheduling depends on spreadsheets and manual logs, resulting in recurring double-bookings, untracked cancellations, and lost academic time. Additionally, basic applications manage conflict verification purely at the application layer, exposing the system to race conditions during simultaneous user bookings.

Summary

The Automated Lab Allocation System is a dynamic web platform engineered to streamline, automate, and optimize laboratory scheduling across educational institutions. By replacing manual, spreadsheet-bound workflows with automated conflict detection, the system maximizes resource utilization, eliminates double-booking, and delivers real-time timetable visibility for administrators, faculty, and students.

1. Introduction & Objectives

Problem Statement Conventional scheduling workflows depend heavily on manual entry, causing frequent timetable overlaps, unrecorded session cancellations, underused laboratory infrastructure, and communication delays during room reassignments or exam seating plans.

Project Objectives

- **Automated Scheduling:** Eliminate manual scheduling errors by deploying algorithmic timetabling.
- **Conflict Resolution:** Identify and block spatial and temporal double-bookings in real time.
- **Role-Based Access Control (RBAC):** Deliver role-specific dashboards tailored for Administrators, Faculty, and Students.
- **Resource Optimization:** Align laboratory assets (such as software packages, GPU capacity, and seating limits) with course requirements.

Project Scope

The project scope covers user authentication, master timetable generation, ad-hoc lab reservations, hardware tracking, and analytical reporting to support institutional decision-making.

2. Key System Modules & Features Admin Module

- Lab Profile Management: Manage lab profiles by defining system counts, installed software, operating systems, and overall capacity.
- Master Timetable Creation: Generate master schedules that map recurring weekly practical sessions to optimal laboratories.
- System Monitoring: Track overall facility usage metrics and process administrative override requests.

Faculty Module

- Ad-Hoc Slot Request: Reserve lab slots for supplementary classes, practical exams, or makeup sessions.
- Resource Filtering: Filter available facilities based on room capacity and specialized equipment requirements.
- Issue Reporting: Report malfunctioning hardware or software errors directly to technical assistants.

Student Module

- Personalized Schedule View: Access personalized timetables detailing assigned venues, workstation allocations, and time slots.
- Live Notifications: Receive real-time alerts for venue reassignments, rescheduled practical sessions, or cancellations.

3. Conclusion

The Automated Lab Allocation System minimizes administrative workload, eliminates scheduling conflicts, and optimizes the utilization of laboratory infrastructure. It delivers a scalable, secure, and intuitive digital framework tailored for modern academic operations.